



Yihuai Zhang

Ph. D. in Intelligent Transportation
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RESEARCH INTERESTS

Theory: Boundary control of distributed parameter systems, stochastic **partial differential equations(PDEs)**, theoretical-guaranteed machine learning methods(**neural operators**), learning-based control for dynamical systems.

Applications: Intelligent transportation systems, traffic flow control.

EDUCATION

City University of Hong Kong <i>Postdoctoral Fellow, Supervisor: Prof. Lu Liu</i>	Mar 2026 – Mar 2027
The Hong Kong University of Science and Technology(Guangzhou) <i>Ph.D. in Intelligent Transportation, Supervisor: Prof. Huan Yu</i>	Sep 2022 – Jan 2026 <i>GPA 4.1/4.3</i>
South China University of Technology <i>M.S. in Vehicle Engineering</i>	Sep 2019 – June 2022 <i>Average score 88/100</i>
Southwest University <i>B.Eng in Vehicle Engineering</i>	Sep 2015 – June 2019 <i>Average score 86/100</i>

PUBLICATIONS

Journal Papers

Yihuai Zhang, Huan Yu, Jean Auriol, Mike Pereira. “Mean-Square Exponential Stabilization of Mixed-Autonomy Traffic PDE System.” **Automatica**, 2024, 170: 111859. (Regular paper)

Yihuai Zhang, Ruiguo Zhong, and Huan Yu. “Mitigating Stop-and-Go Traffic Congestion with Operator Learning.” **Transportation Research Part C: Emerging Technologies**, 2025, 170: 104928.

Yihuai Zhang, Jean, Auriol, Huan Yu. “Operator Learning for Robust Stabilization of Linear Markov-Jumping Hyperbolic PDEs.” **Automatica**, 2026, 185: 112809.

Yihuai Zhang, Huan Yu. “Event-Triggered Control of Congested Mixed-Autonomy Traffic.” **IEEE Transactions on Intelligent Transportation Systems (T-ITS)**, 2026, 27: 1.

K Lv, J Wang, **Yihuai Zhang** et al., “Neural Operators for Adaptive Control of Freeway Traffic.” **Automatica**, 2025, 182: 112553.

Yihuai Zhang, et al., “Low-speed Vehicle Path-Tracking Algorithm Based on Model Predictive Control Using QPKWIK Solver.” **Journal of Dynamic Systems, Measurement, and Control** 143.12 (2021): 121003.

Yihuai Zhang, Yidan Cao, Huan Yu, Lu Liu. “Robust Stabilization of Linear Markov-Jumping Hyperbolic PDEs with Boundary Input Delay”, **IEEE Transactions on Automatic Control (TAC)**, Under Review, 2026.

Z Wang, G Jiang, **Y Zhang**, K H Johansson, A Rikos, “Cooperative Switched Formation Control of Autonomous Vehicles: An Event-triggered Approach to Input Saturation and Time-delay Challenges” **IEEE Transactions on Intelligent Transportation Systems (T-ITS)**, Under Review, 2026.

Yihuai Zhang, Jean, Auriol, Nicolas Espitia, Huan Yu. “Event-Triggered Gain Scheduling of Linear Hyperbolic PDEs via Neural Operators”, **IEEE Transactions on Automatic Control (TAC)**, Under Review, 2026.

Yang Du, **Yihuai Zhang**, Huan Yu. “Event-Triggered Control of Linear Hyperbolic Systems with Markov-Jumping Parameters”, **IEEE Transactions on Automatic Control (TAC)**, Under Review, 2026.

Ziming Wang, **Yihuai Zhang**, et al., “Adaptive Event-Triggered Formation Control of Autonomous Vehicles.” **IEEE Transactions on Intelligent Transportation Systems (T-ITS)**, Under Review, 2025.

Conference Papers

Yihuai Zhang, Ruiguo Zhong, and Huan Yu. “Neural Operators for Boundary Stabilization of Stop-and-Go Traffic.” 6th Annual Learning for Dynamics & Control Conference (L4DC). PMLR, 2024.

Yihuai Zhang, Jean Auriol, and Huan Yu. “Robust Boundary Stabilization of Stochastic Hyperbolic PDEs.” 2024 American Control Conference (ACC). IEEE, 2024.

Yihuai Zhang, and Huan Yu. “Event-Triggered Boundary Control of Mixed-Autonomy Traffic.” 2024 IEEE 63rd Conference on Decision and Control (CDC). IEEE, 2024.

Yihuai Zhang, Jean Auriol, and Huan Yu. “Neural-Operator Control for Traffic Flow Models with Stochastic Demand.” 5th IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE 2025). 2025.

K Lv, J Wang, **Yihuai Zhang** et al., “Neural Operators for Adaptive Control of Traffic Flow Models.” 5th IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE 2025). 2025.

INVITED TALKS

Mitigating Traffic Congestion with Operator Learning , <i>Hosted by Prof. Karl Henrik Johansson</i>	2025
NetCon Seminar of KTH Royal Institute of Technology, Stockholm, Sweden	
Neural-Operator Control for Traffic Flow Models with Stochastic Demand , <i>Oral</i>	2025
5th IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE 2025), Beijing, China	
Neural Operators for Adaptive Control of Traffic Flow Models , <i>Oral</i>	2025
5th IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE 2025), Beijing, China	
Learning for Dynamics and Control of Mobility Systems (III)–Learning-based Traffic Control , <i>Oral</i>	2025
Summer School of CPDE 2025, Beijing, China	
Operator Learning for Robust Stabilization of Linear Markov-Jumping Hyperbolic PDEs , <i>Oral</i>	2025
2025 International Annual Conference on Complex Systems and Intelligent Science (CSIS-IAC 2025), Shenzhen, China	
Event-Triggered Boundary Control of Mixed-Autonomy Traffic , <i>Oral</i>	2024
2024 Conference on Decision and Control (CDC 2024), Milan, Italy	
Neural Operators for Boundary Stabilization of Stop-and-Go Traffic , <i>Poster</i>	2024
6th Annual Learning for Dynamics & Control Conference, Oxford, United Kingdom	
Robust Boundary Stabilization of Stochastic Hyperbolic PDEs , <i>Oral</i>	2024
2024 American Control Conference (ACC), Toronto, Canada	
Mean-Square Exponential Stabilization of Mixed-Autonomy Traffic PDE System , <i>Oral</i>	2024
Joint International Conference on “Automation-Intelligence-Safety” & “International Symposium on Autonomous Systems” (ICAIS&ISAS), Chongqing, China	
Impact of Connected Automated Vehicle Platooning On Mixed Lane-Free Traffic , <i>Poster</i>	2023
The 27th International Conference of Hong Kong Society for Transportation Studies (HKSTS), Hong Kong, China	

TEACHING EXPERIENCE

- INTR 6000G: Traffic Flow Theory** | *Teaching assistant*
Intelligent Transportation Thrust, HKUST(GZ)
Led office hours and gave feedback and comments for weekly readings and in-class discussions. Contributed to the design and grading of assignments.

Fall 2023
- INTR 5300: Nonlinear Control System** | *Teaching assistant*
Intelligent Transportation Thrust, HKUST(GZ)
Led weekly sections and office hours. Contributed to the design and grading of weekly assignments as well as the final exam.

Spring 2024

MENTORSHIP

- Yidan Cao (PhD candidate, CityUHK)
Project: Robust Stabilization of Linear Markov-Jumping Hyperbolic PDEs with Boundary Input Delay (Paper submitted to IEEE-TAC)

Mar 2026 - Mar 2027
- Ziming Wang (Mphil student, HKUST(GZ))
Project: Adaptive Event-Triggered Formation Control of Autonomous Vehicles (Paper submitted to T-ITS)

Aug 2024 - May 2025
- Kaijing Lv (Visiting PhD student, Beijing Institute of Technology)
Project: Neural Operators for Adaptive Control of Freeway Traffic (Paper published in Automatica)

Mar 2024 - Sep 2024
- Ruiguo Zhong (PhD Student, HKUST(GZ))
Project: Mitigating Stop-and-Go Traffic Congestion with Operator Learning (Paper published in TRC)

Sep 2023 - Jan 2024

ACADEMIC SERVICES

- Served as the reviewers of
- Peer-reviewed journals:** Automatica, IEEE Transactions on Automatic Control, System & Control Letters, IEEE Transactions on Intelligent Transportation Systems (T-ITS), IEEE Transactions on Cybernetics, International Journal Of Adaptive Control And Signal Processing, Engineering Applications of Artificial Intelligence, IEEE Transactions on Vehicular Technology
- Peer-reviewed conferences:** American Control Conference (ACC), Conference on Decision and Control (CDC), Learning for Dynamics and Control Conference (L4DC), IFAC Workshop on Control of Systems Governed by Partial Differential Equations (CPDE)

TECHNICAL SKILLS

- English Proficiency:** IELTS (7.0), CET-6 (508)
- Programming Languages:** Python, MATLAB
- Tools and Frameworks:** PyTorch, Tensorflow, Linux

AWARDS & SCHOLARSHIPS

- INTR Outstanding Graduate Student Award (PhD), HKUST(GZ), 2026.
- Grand Prize, INTR Academic Star Gathering-Spotlight Posters, HKUST(GZ), 2025.
- Student Travel Award, IFAC Workshop on Control of Systems Governed by Partial Differential Equations, 2025.
- First-class Awards, South China University of Technology, 2019-2022.
- First-class Awards, Southwest University, 2016-2019.
- Tang Lixin Scholarship, Southwest University, 2019.
- Outstanding Student Leader and Outstanding Student Union Cadre distinctions, Southwest University, 2018.

REFEREES INFORMATION

- Prof. Huan Yu**, HKUST(GZ), Thrust of Intelligent Transportation, Email: huanyu@hkust-gz.edu.cn
- Prof. Jean AURIOL**, Université Paris-Saclay, CNRS, Email: jean.auriol@centralesupelec.fr